ActInf Livestream #017.1 ~ Information flow in context-dependent hierarch

Livestream | Information flow in context-dependent hierarchical Bayesian inference | 1:54:41

hello
and welcome everyone to active inference
lab
to the active inference live stream
today is march 9th
2021 and we are in livestream 17.1
the active inference lab is an
experiment in
online team communication learning and
practice related to
active inference you can find us at the
links and the contact mechanisms here
this is recorded in an archived live
stream so please provide us with
feedback so that we can be
improving on our work all backgrounds
and perspectives are welcome here
and we'll be using good video etiquette
for live streams
muting if we're not talking raising our
hands so that we can hear from everyone
on the stack and using respectful speech
behavior
here we are today in 17.1 coming fresh
off of our
fun quarterly roundtable number one and
the first
of two weeks discussing this paper with
uh the author chris fields and we're
gonna have a ton of time
to get into it today and a follow-up
discussion
next week today our goal is to
discuss and learn this awesome paper and
area

some areas that are probably new to
everybody
and today in 17.1 we're just gonna
introduce ourselves say hello
then chris is gonna give a presentation
and then we'll be able to
ask questions and continue the
discussion from there
so without further ado we can get into
the introductions and warm-ups
so everybody can just go around we'll
introduce ourselves
say hello and then we can also consider
these warm-up questions
so after introductions and anyone
wanting to say something on a warm-up
question we'll go to chris's
presentation so i'm daniel i'm a
post-doc
in california and i'm excited today just
to
hear all of the things that we missed or
messed
in 17.0 and learn and uh
recalibrate and correct and i'll pass it
to
alex
thanks hi everyone
my name is alex watkin i'm in moscow
russia
i'm a researcher in systems management
school and also co-organizer
of active inference lab and they pass it
to scott
thank you alex my name is scott david
i'm the director of the information
risk research initiative at the
university of washington

applied physics lab i'm super excited
about
the the discussion today about
information flows we deal in
as name sounds and risk flows and
information and risk are really
hand in hand so this is going to be
super exciting today and i'll pass it to
siri
uh hi my name is sarah davis i'm a
science artist and actually current grad
student at um in philosophy of science
at university of leipzig
and i'm really looking forward to this
talk today
um who else do we have
donna and chris and then so maybe shawna
first and then chris you can
introduce yourself and take it away hi
my name is shanna and i'm a
mathematician
and um i'll be grad student in the fall
but i've been doing all this stuff and
chris and i are working on a couple cool
things
and i support him anyway hello
cool so chris welcome thanks for joining
us
and we are looking at hello steven so
chris um feel free to just introduce
yourself
and take it away with a presentation
okay thanks daniel thanks for inviting
me to
suggest a paper and join you for this
discussion session
today i wanted to
start with just a quick

presentation to provide some background
context
on this paper which is about context
uh written by myself and and jim
glasbrook
who is a mathematician uh we've been
working together
several years we actually connected and
started working together because we were
both interested in the
perceptual and intentional phenotypes
that one
finds on the autism spectrum and
that's where we encountered uh carl
fristen actually
and the whole active inference framework
in some of his papers on autism
so uh let me just start this little
presentation
and that'll give some background for
discussion
here we go okay as i said just some
some context for uh
this paper about context
and let's see oh okay
so am i so good on on vision and hearing
and all that
looks perfect thank you okay cool
so um here's a standard picture from
carl fiston on the theory of active
inference
uh it describes the interaction between
some system that has internal states and
some system that has
external states and the interaction is
mediated by
sensory states and and active states of
the form of arkham blanket

and when i first encountered this
um it seemed like a very nice
modern statement of ideas that i would
trace
back to robotics initially
in terms of the the
learn slash explore distinction
that one finds in in developmental
robotics especially
and and even from there uh
back to the basic theory in
ethology where you have the approach
avoid distinction
um approach of being very similar to
explore
to acting on the world in order to learn
something
and avoid being
a state in which the organism backs off
and tries to figure out what it did
wrong
so the the theory has a very rich
history and i think
when it's what its current formulation
uh due to fristen and his colleagues
brings to it is a large
body of applicable formalism and in fact
i would characterize the theory of
active inference as it's been presented
as perception and cognition from a
physics perspective
and the presentation
of the theory thus far has been
primarily in the language of
classical physics and
i think that the language of classical
physics
is actually seriously

unsuitable for
talking about active inference it's
complicated
and messy and so
what i'm going to do today is provide a
little bit of background
from a from a more quantum theory
perspective
and i consider today just starting with
a an introductory talk about quantum
theory but i
that won't give us quite the background
that we need for this
so um
i won't be quite so uh
straightforward about quantum theory but
i am i'm happy to answer any questions
about it
and you can go on and on about it as
shauna knows
um so what i really want to
to do is give you another way of looking
at
active inference which i would
characterize as
physics from the perspective of
perception and cognition
so taking the idea of observation
and reformulating physics using that
idea
and you'll recognize from the paper
there's a
a a coco diagram
that's sitting next to something that
looks a lot like a markov blanket
so the cocoon diagram represents the
internal state of some system
and the external state is completely

unspecified
all that's happening out there is some
dynamics
and the job of system a is to
learn as much about system b as it can
in order to make good predictions so
this is still
active inference just from a different
point of view
and looking at it this way it gets us
into
a a cloud of issues that are
all related and i hope uh
by this little introduction to show you
how they're related at least somewhat
give you a sense of how they're related
and
i'll i'll start this this cycle with
system identification
which is the old problem from
engineering and cybernetics
of figuring out what system you're
working with
and in physics this is typically just
taken for granted
and in biology it's often just taken for
granted
um but in uh computing theory for
example
and in what are called device
independent protocols in physics now
it's not taken for granted system
identification is up for grabs
intrinsic contextuality is the topic of
the paper
and it has to do with the unavoidable
uncorrectable effects of
unknown contextual variables on

measurements being made
now that's clearly related to the
question of reference frames
how am i making my measurements
and in fact the first formulation of
contextuality and physics by coca and
specker
was about making measurements with
different kinds of apparatus
and showing that experimental results
could vary even for the same
observable based on how what other
things were measured at the same time
this connects to the frame problem from
ai which is the problem of
of predicting side effects of your
actions
so side effects of making the
measurement in a particular way would be
an example
this gets to the problem of memory if
you're going to predict
side effects you have to remember what
you've done and so
you have to ask about what are the side
effects of recording your memory
someplace because that's a physical
action and finally we get to the
question of ensemble sampling
and the question of what probability
actually means right prior probability
is fundamental to bayesian decision
theory
but it becomes problematic to say what
does prior probability
actually mean and all of these issues
swirl around at least in my view one
issue which is the issue of separability

what does it mean to separate the world
into a set of systems that then interact
and separability
has a very precise meaning in quantum
theory
so this is one reason for diving
immediately into quantum theory
in quantum theory the notion of
sufferability
is defined
in terms of whether a quantum state
factors into two quantum states
so if if i have a system and i can draw
a boundary
somewhere in it that divides it into two
compartments a and b
then if the joint state the state of the
entire system a b factors
into a state of a and a state of b then
the system is separable
and otherwise it's entangled
and entanglement is kind of the natural
state of systems
in quantum theory whereas separability
is the
the only state of systems in classical
physics
so a has a state and b has a state
and we can specify those two states
independently of each other
if this joint system is separate that
means a and b
have identities and we can talk about
their interaction
so i want to go back in history to the
18th century and
and just mention how strange an
idea this is if you think about

physics at the time of laplace
he was mixing the kind of atomism of
democritus
the kind of hard sphere irreducible
particle
atomism with the dynamics of newton
and so laplace's world was a bunch of
atoms each of which was a hard sphere
a monad and they were moving around
according to newton's laws
and newton's laws act instantaneously
so there's no sense in newton of forces
being communicated the forces are just
there and oh this is something that
newton was aware of
and here's a quote from nicholas guyson
who's been very concerned about this
issue of locality and physics and
he in fact wrote a beautiful paper
pointing out that the
passion for locality in physics is very
recent
it comes from einstein einstein invented
this idea that
information travels at a finite speed
before that everything happened
instantaneously so newton understood
that if you go up to the moon and move a
rock
then everybody's weight on earth changes
instantly now that's contextuality
that's a change
here that happens
unpreventably due to
completely unknown events happening
somewhere else
so intrinsic contextuality is closely
linked to non-locality

now in laplace's world of
all these atoms obeying newton's laws
there aren't any boundaries there are no
boundaries around
systems like the earth or you or me
uh or a cannonball so where do those
boundaries come from
in a laplacian world they have to be
imposed from the outside
so we impose a boundary we slice
this picture in two and we say those are
two different systems
now if you think about that in classical
physics
that means that you've created an
infinite potential
that separates the particles from each
other and prevents the newtonian
dynamics from mixing them up
and frank tipler several years ago wrote
a beautiful paper
published in pnas showing that if you
just start with laplacian physics
and you remove these singularities the
boundaries around objects you get
quantum theory
automatically so laplace in a
sense had a theory very much like
quantum theory
and he had a notion of the universe very
much like entanglement who was full of
what einstein would later call
spooky action in the distance
okay so in this picture where we
start with the system and we slice it
down the middle and call the two parts a
and b we're assuming that there's a
boundary there

it's not given to us by the physics
and we're assuming that it stays
constant over time
and in quantum theory unlike classical
physics
the dynamics will automatically erase
this boundary
so quantum dynamics of an isolated
system
is unitary and so it mixes everything
together into a superposition
so if we say that a and b can be
separated
what we're effectively assuming is a
weak local interaction
that's only well defined for a short
period of time
because eventually the dynamics of the
of the
joint system will erase the boundary
that we've put there
okay but if we have
a separable system and if it's finite
dimensional
we have a perfectly general way of
talking about the interaction we can
always write the interaction in this
simple summation form
uh the whole system interaction which
i'll call h
a b with parentheses is the sum of what
a is doing
and what b is doing and then how they're
interacting
and this interaction can always be
written
in the form that i give that i've given
below which is

basically a bunch of thermodynamics
numbers of degrees of freedom
and kt and some measure of efficiency
times a sum of operators and those
operators we can choose a basis where
those operators have
binary outcomes so they question they
correspond to
asking yes no questions which is what
john wheeler postulated the very
foundation of physics was
just answers to yes no questions but we
can always do this in quantum theory
and that's why quantum theory is so nice
it allows us to
[Music]
write down a formalism that explicitly
describes the interaction between two
systems
in terms of exchanging information
and this depends on an insight that
boltzmann had
in the 1870s and it's really this
insight that that
gives us quantum theory boltzmann
realized that
information always costs energy
and what this did was rule out
the way of thinking about measurement
that one finds in classical physics
which is that the observer is completely
detached from the system
and information flows from the system to
the observer
but but that process of observation
doesn't affect the system and boltzmann
realized that this simply wasn't true
observers aren't gods uh if i want to

get
information i need to spend some energy
and that energy flows back into the
world so by getting some information i'm
always affecting what's going on so
from a formal point of view these
operators
the mk implement both perception
and action and that becomes key of
course to understanding active inference
because
active inference is about getting
information and
giving information by acting
okay so to sum this up whenever i have
a bipartite system that's finite and
it's separable
i can write the interaction as
thermodynamics times
questions and that looks just like
classical communication
i can think of my two systems as agents
alice and bob they're always called and
they exchange bit strings
finite bit strings
so now let's think of a specific model
let's suppose that alice and bob are
each
uh alternately preparing and then
measuring
the states of a bunch of independent
cubits quantum bits
and we can think of those as electron
spins or something like that
so um alice in some
time uh prepares in electron spins
and she hands those to bob and bob
measures them

and then bob writes a message by
preparing the electron spins in some way
and hands them to alice and alice
measures them so they've now
exchanged some bits
but there's no requirement that they use
the same z-axis to measure the spins
so if they use exactly the same z-axis
to measure their spins then if alice
writes
zero zero one bob is going to measure
one zero
zero one but if
alice and bob are using different z-axes
to measure their spins
alice will write one zero zero one using
her z
axis and bob will get some probability
distributions over 1 and 0
for each qubit measuring it with his
axis
and the way those probabilities will
look
will depend on how tilted his z-axis is
with respect to analysis
so this is now an interesting situation
and this gets to daniel's question about
shannon information versus other kinds
information with semantics
this is a purely quantum system
there's nothing to introduce any
classical noise
so the information exchange is perfectly
symmetrical
and completely free of classical noise
but by using different reference frames
they introduce quantum noise
and specifically what they lose is the

message
they lose the semantics from the message
so alice encodes four bits bob gets
four bits but the code
may have been lost and the code resides
in the z-axis if they're using the same
z-axis they share a code
if they're not they don't so semantics
depends on
shared reference frames and that's what
shannon information theory
is missing it's missing the idea of the
reference frame that's
used to decode the message
it's just counting bits so shannon
theory is just a quantitative theory of
bits
a way of counting bits and what we're
really interested in is decoding
encoding and decoding so we're really
interested in reference frames
and that will be a central theme it is a
central theme in the paper
okay so sum up every interaction between
finite systems and a sufficient joint
say can be represented in this way
of two agents and they're separated by
a set of bits a set of qubits that set
of qubits can be thought of as a
holographic screen
it functions as a markov blanket and the
information that he codes is completely
specified it's the eigenvalues of the
interaction
written in its hamiltonian form
so we know everything there is to know
about an interaction
in terms of a picture like this and

these pictures are uh everywhere
in biology um everyone is now using
the market mark all blanket concept it's
very common
in computer science exactly the same
concept is called an application
programming interface
and it can be plugged down between two
virtual machines and regulate their
communication
in cosmology the holographic principle
was originally invented to deal with
stretched horizons and black holes so it
separates the exterior from the interior
and it specifies exactly how much
information can be obtained
if you're an observer on the exterior
about the interior
and it's symmetrical so it also
specifies exactly what the interior can
learn about the exterior
but you also see this concept in
particle physics if you're thinking of a
scattering interaction for example
that's mediated by
a vector boson of photon or something
like that
say it's electromagnetic scattering then
all of the information is on that vector
ozone there's no information about
say entanglement on one side that's
communicated to the other side
it's it's just a single value
a momentum transfer for example that's
encoded
by this holographic screen
okay so where do we take this if we have
an observer

and she's got a holographic screen and
she's interacting with
the rest of the universe bob then by
looking at her bit string alice can
learn very little
she doesn't know what z-axis bob is
using so she gets a message but she
doesn't know what code
bob used she doesn't know how big bob's
state space is
so bob could be a small system could be
an enormous system
since she doesn't know the state space
you can't possibly know bob's state
she doesn't she doesn't know whether
bob's state is separable or entangled
so whether there are any divisions in
bob that she has to worry about
uh she doesn't know the dynamics on the
other side so she can't predict what bob
will do next
and all of this is in principle uh
looking out at the world i actually
don't know that there are any separate
objects in it
i can't based on the information that i
can get
at my boundary which is just a set of
interaction eigenvalues
and so these questions have to be
answered some other way they can't
be answered by observation they have to
be answered by a generative model
so hence active inference is based on
what models predict and
what quantum theory tells us about
generative models
is that they're heuristic by definition

they
can't be constructed based on the data
they're
only heuristics so where did they come
from
and here when we look back at our
picture our original picture of a joint
system being divided
uh recall that this boundary is
arbitrary
and it doesn't last very long
so this joint state isn't separable for
very long
and the only asymptotic dynamics the
only dynamics that's long lasting is the
joint dynamics
so alice's generative model
which is implemented by her dynamics is
effectively a sample of the joint
dynamics
it's just a temporary sample of what
would be going on
anyway if there was no division
so it makes sense that alice's
generative model
contains information about the world but
that information
did not come from observation it came
from
the time before the boundary was imposed
and it will still be there after the
boundary goes away
okay another thing that quantum theory
adds to the acting inference picture
is that it recognizes that all of the
energy that's required to run the
calculations
also has to come through the markov

blanket
so if you think about the states
on a holographic screen a markov blanket
of any information encoding
if that interface is the only interface
between the two systems which is how
we've defined it here
then some fraction of those bits can't
be informative because they have to be
burned as fuel so by definition they're
meaningless
now this tells us again something very
deep which is that
the context of a measurement of can
never be specified
completely and that again is in
principle
because some of the information on the
blanket has to be burned as fuel
and we see this on the output side too
of course
of some of my output
on the world some of my action on the
world
is just dissipating what for me is waste
heat
and i have no control over that that's
part of my metabolism
but i'm side affecting the world all the
time by
dissipating this waste heat which i
don't
think of as an action but which is an
action from the point of view of the
physics
so here again we have an addition to
the theory that we get from thinking
about the dynamics from a quantum

physics point of view instead of a
classical physics point of view
okay so this is the picture i showed
before
this set of qubits is the markov blanket
of generated model is represented by
this
coconut diagram and
it becomes clear of what the generative
model is it's a combination of a
reference frame
and some combination some computation
and it's clear what it does it
specifies the option space for
both perception and action it encodes
everything that's detectable
in everything that's actionable about b
4a so here just a couple of cartoons
for example if alice looks out in the
world and what she sees
is a red rectangle interacting with a
green diamond by some
interaction then she must have reference
frames for
seeing rectangles and diamonds and she
must have
state observables for red and green and
she must have a
memory so that she can keep track of
what's going on with those state
observables long enough to construct
this picture of an interaction
so all of the semantics is inside the
observer it has to be provided by these
structures
inside the observable observer so the
semantics that's imposed on the world
is imposed it comes from the generative

model
it's not out there in any sense
part of of what the observer has to do
is write
memories if the observer is going to see
change and since memories are classical
data those memories are effectively
written back in the blanket because the
blanket is the only place where any
classical data live
so we can we can infer from this that if
a memory looks internal to a system
that system has to have
intercompartmental boundaries inside it
to to write those memories on so this
actually tells us something gives us an
expectation of physiology
in organisms or in any sort of
artificial structure that
memories have to be written on
boundaries of some sort
so again here's this this swirl of
issues
that all surround this question of
separability
of the world being separated into
distinct
pieces and i've mentioned all of them
now except the frame problem
which is the problem of predicting side
effects
and clearly if you don't have
access to the dynamics outside you
and you don't have access to all of your
actions
because some of them are just heat
dissipation
you can't possibly predict side effects

all side effects so the frame problem is
unsolvable in principle
in this setting
so in this paper what gemini set out to
do
was to construct a general
representation of
these reference frames and because
they're physically implemented they're
quantum reference frames they're not
just
abstracted and
these things specify both what the agent
can expect
and what the agent could do so they
completely define the option space
for active inference for the agent so by
having a general representation of the
agent's reference frames we know
in a sense everything interesting to
know about the agent
then we wanted to distinguish the
reference frames that can be used
simultaneously and these are the
co-deployable reference frames from
those that can't
and the distinction again is given to us
by quantum theory it's just
operator commutativity so for example i
can't
deploy position and momentum at the same
time because
those operators don't commute
they side affect each other it's the
other way to think of it
and physics is full of operators that
don't commute
so my my action

space is is not unitary it's
segmented into uh sets of things that i
can't do at the same time
without introducing contextuality
so we showed that this
non-co-deployability of reference frames
actually induces intrinsic contextuality
as it's defined in the literature
what that means is that it creates
undefined joint probability
distributions
um not just probability distributions
that are difficult to make sense of
or that needs some information added to
them
but probability distributions that
actually can't be consistently defined
why not because there's the information
that would have to be specified
to make them well behaved as information
that you can't have
that you can't get even in principle
and then our final objective was to tie
this explicitly to the frame problem
which is
a canonical problem that artificial
intelligence
discovered by mccarthy and hayes back in
1969
the unpredictability of side effects
so that's that's it for my introduction
kind of context setting
and i am happy to
um as soon as i can get to this button
again
switch to discussion
chris thank you so much for that there
it is

yep we got it now thanks so much for
that
really amazing presentation i'm sure
everybody was having some pretty
interesting thoughts from what you were
communicating really though thanks again
for coming on the stream
so um for those who are here please
write down your questions
and then when you ask your question
let's stop after the question so that we
can get a response to that
question and not continue on with the
question and then we'll
raise our hands to hear from everyone so
i'm gonna first ask a question from the
chat and then stephen i saw you raise
your hand so anyone else
raise your hand otherwise there's a
couple questions from the chat
so here's the first question from the
chat can the context-based paradigm
especially quantum contextuality help to
answer
how does spontaneously generated thought
arise in the brain
if yes how how do you see this relating
to the brain
um good question
um i
actually suspect that
all cells are quantum computers
and that to to really get a handle on
information processing in cells
we'll eventually have to use quantum
information theory
and the the reason for that is that
cellular energy budgets are really

small and and place a pretty strict
upper a strict upper limit
on how much classical information cells
can encode
and that upper limit is much
much smaller than the information
required to encode
things like protein confirmation
so i think that that cells are
in fact uh devices where
contextuality and entanglement play a
role
now what what spontaneous thought means
i'm not sure oh
i'm um
very sympathetic to
the idea that thoughts
are and emotions and and everything else
that we experience
[Music]
just the experience tip of a very large
iceberg that we don't experience
so i think our experiential
access to
what our minds are doing is extremely
limited
and um
so how how what we actually experience
as a as it were rises to the surface is
uh a good question
i tend to think that global workspace
type theories
are a good classical answer to that
but all of those ideas
i think will eventually need to be put
in a
much better mathematical framework
thank you for the answer that's a long

way of saying i don't know
but it's what spontaneously came to mind
so how could we ask for anything else
stephen and then anyone else who wants
to raise their hand
yeah thank you um i was just curious um
you mentioned
sort of the quantum noise and the the
the way that if you um have
the z-axis um that you if you don't have
the z-axis you can't decode
back the information and often with
quantum
effects it's like it's when you have
very low
uh degrees of freedom for like a
molecule or something that you
particularly can see it
compared to you know when it's in the
melor of many degrees of freedom
so i'm wondering if there's like a um
certain scales where like
the the interactions are in the cell or
molecules are very quantum
like and then it becomes more contextual
and a bit more blurry and then another
scale it becomes a
bit more quantum like maybe once you
start to get to brainwaves interacting
and is the contextuality piece
moving it in a way it's another way of
moving it in and out of a more classical
um kind of noisy space
um so that it's like where it's got more
um
noise and it's less easy to see sort of
more distinct quantum effects
so i was wondering like is there

particular scales at which
you see the quantum effect being more
like you think a traditional quantum
like flipping between states
um and if there's other scales where
it's much more
like fristen talks about this kind of
non-linear
miller
great
yeah that's that's a very good question
is the question about
is there a quantum to classical
transition and if so where
and
that that question is typically posed in
terms of decoherence
and in in
biological systems um
estimates of decoherence times are
anywhere from
femtoseconds down
there's the famous paper by tegmark from
2000
uh criticizing the hammer off penrose
theory where
um his argument against hammer off and
penrose is basically the d coherence
happen many orders of magnitude faster
than
than any cognitive processes or any
neuronal processes
so based on that type of reasoning
biochemistry is thought of as being
classical
somewhere above the say finto second
type of time scale
so molecular dynamics calculations

are basically classical when they're in
like finto second types of time steps
and that introduces the question of what
is noise
right what what does it mean to say that
there's thermal noise in a system
um when we think about thermal noise
classically for example we might think
about a protein
that's sitting in a bath of
water and the water molecules are
thermally excited so they're banging
into the protein like
little billiard balls and that's in fact
one way of
thinking about a decoherence model
so what we've introduced here is
this idea of a temporal sample
that we're looking at the protein
molecule for a long enough period
that the water molecules have had a
chance to bang
into it from many different directions
and we're somehow averaging over that
temporal sample
so that gets to this question of
ergodicity that came up in the earlier
discussion
and it it makes us ask how
am i implementing this this temporal
sample and
the the obvious answer is i'm
implementing this temporal temporal
sample by
measuring the state of the protein in a
particular way so i'm i'm using
an enzyme say to to measure the state of
the protein

and the the enzymatic reaction has some
particular time constant
so it takes a certain amount of time
for the enzymatic reaction to take place
and during that time the protein may be
wiggling around
so i think that that all of these
questions about classicality
in the end come down to questions about
measurement
and the temporal resolution of
measurements
implemented by something an instrument
or another part of the biological system
or an observer of some kind
so this is a this is kind of a long way
of saying
that i suspect that this question of
classicality is
always a question of observation
and the the sort of coin in picture that
that i outlined implies that
the classical information only exists
on the mark on blanket that there's no
classical information anywhere else in
the system
so classical information may not
in a sense be ontological at all
there may be no ontological classicality
uh if this kind of communication picture
is correct
now if that's the case noise is
um epistemological
noise is an artifact of making the
observations in a certain way
with a certain reference frame right an
instrument is just a reference frame
thank you for the deep answer chris

we're gonna go
blue scott and then a question from the
chat
hi chris thanks for the um
contextualization
talk that was really nice um so
i just was wondering you just mentioned
earlier like the prospect
of each cell being a quantum computer
and i'm also like i have a similar kind
of thought or
theory and i just was wondering if
you've heard of like quantum cognition
and the positive molecule with the idea
of these
entangled pyrophosphates as as holding
kind of quantum cognition cellularly
have you heard this theory and
do you think it's credible or do you
have other
suppositions as to where this quantum
cognition could occur
intracellularly
i've certainly
heard and seen various studies of
localized quantum effects
in particular molecular structures
so
stacked pi domains and and so on and
there's the whole story about
uh is there quantum coherence in
um um
goodness i've just dropped the word
light harvesting molecules or
electron transport or something like
that
and um you know there's the
hemorrhage theory that there's that

there's quantum processing in
microtubules
and in all of these approaches
one is thinking about a quantum
process happening in a background
of a classical cell once one's thinking
the rest of the of the biochemical
situation as being classical a classical
environment of some kind
in which this single quantum process is
happening
and as i sort of expressed in
in answering the last question i
i suspect that that entire way of
thinking is
probably wrong
that we we can't actually think of
bulk biochemistry as being classical
so um trying to think of
a particular event as being
a quantum event against a classical
background
may just be misleading us about what's
how information processing is actually
occurring
and if we think of of the interior of a
cell
say the entire interior of something
like a bacterial cell
as being in one coherent state
then um it's not that
that their particular quantum molecular
states
is it's that the entire interior is one
big entangled mess
that demands a very different way of
thinking about biochemistry and
i don't even know how to formulate

that way of thinking about biochemistry
but i suspect that approaching it as a
as a quantum information
problem where we're extremely
explicit about where measurements are
made
would be what would have to be done
thanks chris really amazing about the
relational
and the semantic biochemistry that's not
how we got it in 103.
so scott and then a question from the
chat
chris thank you that's a absolutely
fascinating presentation
a number of questions but i'll just ask
one or two
um one of the challenges of quantum
physics
or challenges to quantum physics being
applied in the macro level
is that you know don't take people have
said don't take the concepts and apply
them to social constructions and things
like that
but here the um origin of the
active inference notions or the
relationship to the acupuncturist
suggests to me um the opportunity
to look at phenomenon of much larger
systems larger scale
social scale um and so
i wonder about your um uh acceptance or
enthusiasm for that notion of
applying these ideas
to uh social systems political systems
economic systems other systems that have
interactions that can be described with

active inference
um and this feels to me like a nice way
to say hey
we're talking about quantum politics
quantum
markets we're talking about uh prionic
governance with conformational changes
of narratives
um and so a lot of the work that we've
been doing on the information
side is gonna is i've been saying
without knowing you've explained to me
what i've been trying to say for years
now
in this talk and i've been saying
without
knowing why that data plus meaning
equals information
and the meaning part is what you just
explained to me what i meant by meaning
and i always thought of it i was a
lawyer for 30 years so i always thought
of it as the meaning like a contract
with an enforceable
narrative which is meaning but you've
just given me a new way of looking at
that meaning from a quantum perspective
which is extremely robust
and scale independent i just wonder
about your comments on that and your
enthusiasm for that kind of application
thanks oh thank you that's
that's a very interesting observation
um
i think the experiments are going to be
very difficult
but i think if we can figure out how to
do the experiments

we're going to find contextuality and
entanglement
kinds of effects all over the place
and i would i would point you to the
work of
of uh atamar zad zafarov and his
colleagues some of which
is referenced in the paper
in in their formulation called
contextuality by default
what they've done is
translate this kind of most standard
model of
quantum contextuality into classical
probability theory
and and given this criterion of
if a classical probability a consistent
classical probability
distribution cannot be defined over a
set of data
then it displays contextuality
and they they published a
series of papers in the past four or
five years
on this approach and one of them
deconstructs a large number of claims of
quantum effects in psychological data
which have been reported now by numerous
groups around the world
with various kinds of experimental
protocols
and there's active controversy between
members of these different groups about
what is and is not evidence of quantum
behavior
so it's an enormous literature now
but using the contextuality by default
approach

zafrov and colleagues
his student cervantes is one of the lead
people
who's done this work have
constructed extremely careful
experiments
where under their their best criteria
there are robust intrinsic contextuality
effects
and interestingly they have to do with
meaning
so one that we refer to in the uh
in the paper is called the snow queen
experiment and it's based on
uh interpretations of the hans christian
andersen story
so i think that as
as the tools and techniques for
developing these these kinds of
experimental protocols are improved
uh we're going to see more and more uh
effects that are
very very difficult to challenge from a
statistical point of view
i mean the problem is as therefore often
and i'll point out
is that in physics if i'm doing
something like a bell experiment
i can put my two observers far enough
apart that they can't communicate
just due to the speed of light in the
time it takes to do the experiment
and you can't do that in biological
systems
so you always have this communication
loophole
to deal with and what what zafrovan and
colleagues are really trying to do is

develop the statistical analysis
techniques
uh to get around the communication
loophole in an arbitrary experiment
you know carried out with college
undergraduates
and i think they've been successful
thanks for the response so we're gonna
go
to a question in the chat and then if
anyone here
who hasn't raised their hand yet raises
their hand otherwise we'll go back to um
someone who has
so this is from sudakar in the chat how
will an
agent acquire information for their
generative model
you've uh said that there has to be some
heuristic to infer
the information even from the joint
world and you said that the object
semantics have to be inside the observer
but how does the observer
learn that in the first place for their
generative model
that's that's an excellent question and
again i the only answer i can give is i
don't know
i'll but i think that i think that
biology gives us
hints
if you look at the process of cell
which is one of the best examples of of
separation that we
have to deal with that we can really
manipulate in the laboratory
then and if you ask how much information

is
is shared between the daughter cells
well clearly their genomes are shared
but the genome is only a very tiny part
of the shared information
they also share information that's
encoded in their cytoskeletons
they share information that's encoded in
the cell membrane
they share information that's encoded in
the cytoplasm itself
they may share aspects of their
bioelectric states
so there's a lot of information
that is inherited if you will across
cell division
and most of it has not been quantitated
in in any reasonable way i mean it's
easy to quantitate the information in
dna or protein
but it's extremely difficult to
quantitate this other information
and i'll just give you one example of
that's from my colleague mike levin's
lab at
tufts they work on regeneration and
planaria
and they've been able to show um
experimentally
that it's been known it's been known for
quite some time that if you
alter the
ability of cells to communicate
bioelectrically
then you can get planaria to regenerate
with two heads instead of one
but what mike's lab was able to show
is that this is epigenetically heritable

so planaria that looked perfectly normal
may have inherited the tendency to
regenerate with two heads after injury
and this happens without changing dna
without changing rna without changing
protein content without changing any of
the kind of obviously
inheritable factors
and have been able to show
experimentally that the bioelectric
effect happens
first before the observable changes in
gene expression
so here's a case where a completely
uh unlooked form kind of information
is epigenetically inheritable and makes
a huge difference to what the animal
does
with his with his physiology with its
anatomy i mean the the two-headed
planaria have two perfectly good
operational brains that are
independently able to direct behavior
uh so they're they're pretty weird
creatures
thanks for the response and just briefly
that's one reason why the information is
in the genome kind of zoolander
take doesn't work well because that's
assuming there's this map
between the source code in the program
but then there's these alternate stable
states that exist
because the semantic or the epigenetic
or the environmental frame
so it's not quite as simple as just this
one mapping that you made
so that's a really nice point all right

we're gonna have
sarah and then shauna and then a
question from chat
thank you go ahead sarah yeah i
um i have almost like two different
questions or directions and i'm not sure
how they're gonna match so i'll just
start with one and see where it goes
um one it seems to me like the
the the way you frame the quantum um set
up you know you have a you have b
you could actually kind of do this with
any problem it's probabilistic type
notation that you're using but
you know you have a you have b they have
the joint distribution of the two
and um it seems like
i i guess the question that comes to my
mind is like what happens
when we don't if if we were
hypothetically because i know our world
is constructed around reductionism but
if we were hypothetically able to
separate out
or to not think of things as um
reductionist you know and so like for
example
um how would our how could our math
possibly be different if
we were looking at things as only
spanning sets rather than eigenvalues
and eigenvectors
um yeah so this is kind of where my mind
goes is like
is there a is there even a kind of a way
to approach a problem
purely through contextuality rather than
than you know reductionism of any kind

that's yeah
hmm
i don't know on the on the math front i
may actually defer to shauna here
because
as she she does know much more category
theory than i do and i suspect the
answer is a category theoretic type of
answer
but i'll make one comment which is that
reductionism is one of these complicated
words that has many different meanings
for many different people
but one way to think about
what we're doing in quantum theory is
merely labeling
and we're labeling things for
convenience with the full
understanding that our labels
in a sense don't have any physical
meaning
they they only have meaning
in the context of particular
observational outcomes
that we've made by measuring things in a
certain way
but that in itself is a problematic
statement right and this is
this is in a sense the core of the
copenhagen interpretation it's
it's what bohr talked about in his
paper in nature in 1928.
uh he basically said at the end of the
day
we have to talk about apparatus and we
have to talk about
colleagues and we have to talk about
classical communication

and in a sense quantum theory tells us
that
all of that is um
a poor approximation
but it's the only language that we've
got
so uh i i think this this
question of of how far can we go
without uh reifying
separation is a very good question
and as usual my answer is i don't know
awesome thanks for the response you have
any thoughts yeah let's go
oh yeah i wonder if i should um ask my
two
yeah category theory is really powerful
and when chris says there's only one
logic no no no he and i are trying to
find another one
something called like a chromatic types
and i was actually going to ask you
chris if you think
you know the phenomena of regeneration i
think could actually be modeled by when
we're trying to do that chromatic
filtration
right it seems like regeneration seems
to be some kind of um
morphism equivalent of when the stop
mechanism is what does it mean to stop
maybe that means stop means
not making more morphisms and the two
had a panera like actually has a two
category structure
right that has a one morphism and a two
morphism and i think that's what's
controlling the central nervous system
right that it if you can have two

morphisms and you could have this double
head kind of thing
and so whatever the stop mechanism is i
think it's almost like
remember when i was showing you those
filtrations how you could actually
make more equivalences and more
equivalences using that boost field
localization
so i think you know there's another one
called the topological localization
where the objects and the categories are
reflections and that might be
what might help um sar a little bit is
that there's
different categorical models for
interaction anything interaction based
should be immediately category here i
think the tensor network step
fails because like a node is just a
point
but in category theory you can add space
to that point
by making the point something else and
then you're working in this modulized
space
so that was my thought that about like
spawn and you know when you were talking
about spontaneous thoughts whoever had
the question about that
i mean i also think that like our work
on the pro finite condition chris could
also account for like
thought whatever that is right because
um
the thought doesn't really last right
voice speech doesn't last the speech
fades based on the composition of the

molecules in the air but something
remains i mean we all
continue it seems to be that holographic
copy of the pro finitely mini right in
the diamonds and stuff
so i just want so for syria you know i
think a bunch of this category theory
could really help you with uh
you know uh you know it's like you have
this object and topology called like a
sheaf which is this very
very complex structure but it attaches
to a topological space and it's like a
little bag and it keeps track of all the
data
you know and uh so if each space has
these structures attached to it that
keep track of what happens locally you
can look at the modulized space and the
isomorphism classes of those little bags
so what's happening in the isomorphism
space with the little bags probably
seems to be
what this like global consciousness
things is
so you can upgrade that one dimension
using category theory you can work in a
stack
and a stack is going to be a sheath that
doesn't take values in sets anymore
all of quantum theory has taken place in
sets and i think it needs to be a little
bulldozed like it needs to go
up if you can take values in categories
then
in each point you have so much more
space to move around
right so then if you actually have this

object that takes values and categories
and not in sets then you've
allowed yourself more space so that you
could have these
phenomena like the the planera and stuff
like that but if you're just working in
like
uh like points and like dots and like
nodes connecting i think it's just not
enough but you can do all these other
things
but um i'm not sure if i answered
anything but yeah chris i think that
regener i was going to ask if you
thought would
like our which we need to pick back up
the chromatic you know our chromatic
type over uh
temporal logic if that could actually
work with this
i'm fascinated by the regeneration and
the stop mechanism which is weird
you know yeah i i'll just quote
mike levin telling killing me
over and over again and everyone else
who will listen that the most important
question of biology is when to stop
yes now how how how does
how does the embryo know that it's done
now i can stop i'm an adult
no one knows and that the question even
isn't even asked that way
but when you think about it it becomes
it is a key question
because it's it's related when you said
like uh our experiential
phenomenon uh capacity is just
this you know the the tip of the

non-experience so it's almost like
if the stop is the iceberg of everything
else i'd like to see what's the
everything else
you know
thanks for these points it's really an
amazing
mapping that is rigorous and something
that we should be learning more about
how the point
can be just the tip of the iceberg kind
of where it's piercing
something that's a lower dimensionality
and so
what do we do with that so i'm going to
go to a question from chat and then
can i ask can i ask a second part of the
question i was working with
go for it it's this is and feel free to
just be like
eh it's not great just just move you
know not answer it if it but i
i can't help but analogize
um something that i just yeah i cannot
ever let go which is
um in electrical engineering
or in electronics essentially you know
you have this
well in anything but i i really like
electronics because it's very tangible
you have the real component and the
imaginary component of
of of signal you know that goes across
an interface and so i'm always really
fascinated by that because it always
seems like a really good analogy to this
thing that we can look at through math
or quantum mechanics but with

electronics you actually have
you know a thing that you can put on
your desk um
because you could think of this this
this this phase change that happens at
interfaces
as the context and and so i've always
been interested
in in trying to look at that thing that
gets lost without looking at the
at the current or the voltage like
measuring at a point but actually just
keeping track of the context as things
move across interfaces in a circuit
it's not even a question i apologize for
that but if it brings up any thoughts
great
either of you two if you'd like to give
a thought otherwise
it stands and i love how many ideas
people are bringing up this will
definitely be an episode for people to
re-listen to and make some knowledge
network stuff
uh all i'll say is phase is extremely
important
this is important it's a very it's like
phase according to whom
so chris and i really torn apart this
reference frame idea like according to
whom
at what point if phase was like
non-local then everything would be the
same thing
but so what point do you actually have
you know electronics you assume that you
have a
free that you're like local and stuff

but
um electronics are fascinating it's
still like kind of tangible but if you
work on that
that space that's pre-actualization
i think it's very just like for those
presented it's very very difficult to
say
who you are what the boundary actually
is and if the boundary
um like a boundary in mathematics is
like an asymptote or something right
so it's like a horizon or a koshi
horizon or something like that
so inside that very notion of a boundary
you already have
causality and then this you know tensor
product and there's all kinds of other
tensor part there's wedge products and
you know it's it's i think it's really
hard to say what a phase is
without bringing into play all these
other reference frames that are um
so fragile cool
i'm going to do a question from the chat
and then i have a simple question
so a question from the chat is do you
think a reformulation of quantum
mechanics is necessary
in order to tackle the problem of
consciousness or life
from a full mathematical generality i'm
mainly thinking of the notion of
intrinsic information from
tishbe's paper information theory of
decisions and actions
but also maybe other approaches like
self-organized criticality

more generally i'm curious how necessary
is it to bridge our understanding of
thermodynamics information theory and
biology or control theory
to have a good understanding how far
along are we and are there any concrete
problems for a mathematician
i read what they say you know that's
that's an omnibus
question i i guess my my first answer
would be
yes uh we do need to bridge all those
all those disciplines uh in order to
make progress
um whether
it will tell us something deep
about what experience
is in some ontological sense
i have my doubts about uh
whether it will tell us something
interesting about
what experience um
in a sense does how it works
uh i think it may tell us something very
important about that
but yes i i i do think that these
disciplines
need to be bridged i i think we're
a bit of an impulse that's been imposed
on us by the disciplinary structure of
science
thanks for the answer i'm gonna ask my
simple question
and then we're gonna go to stephen and
blue and also sarah i know
you had some other kinds of questions
that were a little bit less about the
math but

here's my simple question where does
active inference
come into play or for those who are just
learning about active inference and
curious to
how it might be the bridge for all these
different areas
um what do you think active inference is
doing here
well i as i said at the very beginning i
i
think of active inference in terms of
this
fundamental decision that that every
system has to make all of the time
between of
do i explore do i act
or do i retreat and
and and try to do a kind of post-mortem
on just
what just happened and learned something
so do i modify my priors
or do i try to modify my my experience
and
i think thinking of active inference in
terms
of
what reference frames are being deployed
it just allows us to pose that question
a little bit
differently it allows us to think of the
question
in the form of
do i make another measurement do i act
on the world
or do i
modify my reference frame
right you can think of a reference frame

uh like a category
like a cognitive category so we have
cognitive categories like this is a cat
and that's a dog
and this is a table and this is another
person
so modifying a reference frame
can be thought of as going into your
apparatus and
changing the way it works you can also
think of it as
modifying my category modifying the
things i'm capable of seeing
and i i think you asked in the in the
previous discussion last week
what does the theory say about the
the capabilities of organisms and or
does it does it put a limit of what
organisms can do
and i think in a sense that's the
question we're trying to answer
to to what extent are
you organisms have flexible capabilities
to what extent can they retool their
cognitive systems
even if it's even if it's just e coli to
what extent can
can something like a microbial
population retool what it's capable of
and again biology gives us these
beautiful examples
in terms of things like lateral gene
transfer where
uh microbes in a community can pass
around genes and so can
can change very radically what they're
capable of doing
so i mean from an active inference point

of view
that's a question do i
do i keep going with the capabilities
that i've got or do i go shopping for
some new dna
cool it's like the active turn
we don't know how far or what else what
other stones are going to be turned
when we actually start thinking and
doing instead of
just thinking about doing so we have
stephen then blue then scott
very excited to hear you talking about
the work of my mike levin because
his work's very uh when i saw the video
of his work recently like last year was
like wow
it started to bring the whole idea of
swarming and how swarming
actually is underpinning a lot of what
happens and the the
the work with bioelectrics is really
interesting and i
i've been interested in how if that
bioelectric aspect seems more maybe
more closely tied to the sort of quantum
transitions um
and um just maybe a little clarity on a
couple of things because often
traditionally let's just ask the
question and clarify it
okay um so the question is um
when you have these quantum effects you
know normally they may be just you think
of things flipping between one state and
others another state in more sort of
basic chemistry thinking about like
energy levels or something like that but

you have many levels potentially so is
it a case that
you're able to still infer
from the noise what's going on with
quantum states
and therefore this gives us a way to not
have to just
go for like the attractor theory and
complexity or catastrophe theory or
phase theory that we use at the moment
but you're able to still
extract quantum state information
in ways that's not just classical like
one energy level to another energy level
but you get this more
higher information context
yeah good question i i think what you're
really asking is
uh where are the experimental tools that
let us
uh probe this domain
and um i wish
i knew and this is something that mike
and i talk about for example
how could we go about looking for
contextuality effects for example in
things like bioelectric effects and worm
regeneration
and i don't know i think i think there's
a huge opportunity space here for
for people to think about uh
ways of doing experiments that we can't
experiments that we can't do now
or have never thought about being able
to do
that
probe a domain of correlation
that is non-classical or super

determined or however
however you want to think about it
and i think that the the more we can
think of biological systems in
in communication theoretic
terms the better prepared will be
to design experiments to look for
what are really communication theoretic
effects and quantum theory is really
about
communication
thank you cool blue and then scott
so first because shanna dropped off i
don't know if she's gone or if she's
gonna rejoin and it's kind of a
follow-up to some of the stuff that she
was
saying but i can still ask it um but let
me pass to
scott and then i'll go we'll we'll go to
scott and hopefully um
we can have some future category streams
to really learn more about
this area so yeah scott and then back
split
great um thank you this is my
smile is hurting my face it's so such a
great discussion
um the two things i want to talk about
embodiment and this is
related and one is social level
experiments so
the first one is embodiment it seems
like we have these internal states or
embodiments of
prior external states or their learned
states
and so it's kind of interesting because

the external states in the internal
states have bearish
significant relationship through time
not just through that individual
organism's markov blanket but by the
inherited
and other factors that are become the
priors
right and so it's kind of interesting
because they have embodiment externally
and internally
maybe that's extremely fluid right
because if you talk about systems at
different levels they're different
types of systems cell systems social
systems organisms organizations
they all have different affordances and
their embodiments are not necessarily
they can be um
they're not necessarily inconsistent
when they're different right you can
have an embodiment of something
uh if i eat a lot of fatty food i become
fat but that doesn't you know
anyway that's one thing is the
embodiment if you talk a little bit
about that
for me from my perspective if you're
looking at risk and change when you
information being shifted around as a
lawyer ex-lawyer i look at
risk and risk being moved around and so
one of the things that seems to me when
you have the embodiment question
is that among the relationships then you
get the question since entropy or
disorder is going to increase
universally

what we're doing is we're um managing
local meg entropy in these systems right
we're
sharing uh local entropy and so the
question is
is that dynamic something that um tends
to dissipate
the markov blanket as a standalone thing
maybe markov blankets
are a general thing that's one question
okay let's get that
let's try to do one question because
it's really you know in a
body domain let's just try to yeah
thanks for the really great question
go ahead yes okay um
this is a wonderful question i'll make
uh two comments
one is in terms of of reference frames
which i always come back to um
i think what what what quantum theory
contributed to the
to the notion of a reference frame is
it insisted that we regard reference
frames as
embodied as physically implemented
and stop thinking of them as merely
abstractions
and we can we can go back for example to
the idea of
cognitive categories you know my
my idea of a table is
not an abstraction it's a
it's something that's implemented in my
brain
and it can be changed
by operating on my brain
and i can change it that way we don't

know how that works
right but but what the
what the the embodied movement or the
the 4e
framework whatever what that forces
cognitive science to think
to think about is these are not
abstractions we're talking about things
that have been implemented
so i think that's an incredibly
important point
um the second point i want to make is
about
boundaries and i showed that picture of
of
uh right alice looking at her at her
holographic screen and listed the things
that she doesn't know
but i could have added to that that
alice doesn't know
where her boundary is
if if i'm a system
and i'm i'm looking out at the world
through my markov blanket or through my
my encoded screen then
what i'm seeing is this array of
information that's painted on my screen
right this array of bits
and and fristen's point and the point of
the holographic principle is
i don't see what's on the other side
but you can flip that around it's always
symmetrical i don't see what's on my
side
either and
that's that's why i tried to make this
point very briefly about
memory how if we think of memory as

classical
then it's got to be written on the
screen because that's the only focus of
any classical information
so if i'm looking at my screen the thing
that i
don't see is myself
and since i don't see myself and i don't
see the other side i just see the screen
i have no idea where it is
so when you when you raise this question
of
a social system or or any other kind of
complex system
um we as as
observers of that system are using
our reference frames or our cognitive
categories or
our however you want to describe them to
draw a boundary around that thing
that boundary is our imposition
it's not the system's imposition the
system doesn't know where its boundary
is
is there a talk is there a tautology in
there though because that's
interesting right because we're using
our system we're drawing the system
within
our boundary it seems like that's an
intrinsic quality of the thing that
system can
can a system know it's open kind of
system know others and you can only know
it through the boundary
but you so you can't know yourself
because you're not written on the
boundary so it's kind of it

well let me let me get to the next
question because it relates to that so
the next question
but the quick answer to that question is
yes yeah
i like that's a good question it's good
when there's a binary answer
so the next question was i was
wondering if we can um if the
for experimentation this is the question
of answering experimentation
if and to the extent that this is
there's scale independence and active
inference and so that it applies at cell
levels and social
etc why can't we look at the social
level to run experiments
and then apply those what we come out
with at
other levels so it's similar to when you
do a qubit at a macro you know they do
cubit
qubits that are manifested in different
ways in different kinds of systems you
know ionic qubits and things like that
so you can measure them in different
ways
so we could be observers of the system
to
at the social level to identify some
qualities and the reason i'm asserting
that as a lawyer
even though the social has a lot of
variables what we do is as lawyers is
create
fewer variables right we take a meaning
thing we say hey
everyone in this area of the statute has

to behave this way
and so then internally everyone can
absorb and embody the idea that
i know everyone's going to stop at a red
light i know i'm going to stop at a red
light
that's the story of the red light red
means stop you're taught it as a kid
so it becomes instantiated and kind of
axiomatic
so one of the things about the law is
that i'm going at it from a slightly
different way of
effectively going in and trying to
create
non-variables in the markov blanket that
are shared
right and so and then so i wonder about
the embodiment
relates in a way i think to the
experimentation question
because can we do let's call them
artificial embodiments
into the internal states through things
like law or folkways or shared meaning
or norms or things like that
that we can then experiment against is
what i'm wanting cross-cultural
experiments things like that
that then may have application at other
levels
i i think that's i think that's a
fascinating thing to try to follow up on
and um
i would say one of my one of the things
that seems most mysterious to me
is language is is the very possibility
of classical communication of what

what you're talking about a physicist
would call super determinism
and super determinism
is essentially the same thing as
entanglement
if if i don't have a choice
about what reference frames to deploy
if there's a correlation in advance
super determined
uh by the past between the reference
frames i use and the reference frames
you use
then and in an important sense we're not
independent systems anymore
right we're we're entangled but that's
what language
is in
in some sense and so
i i i agree with you we should be able
to
redescribe
communication generally in these terms
of of
reference frame sharing reference rank
choice of
super determinism
and apply some of this math that's been
worked out in other places
to reinterpret communication
and i i'll mention a paper by a guy
named
alexi grinbaum it's a philosopher of
science at saclay
and it's a paper on uh device
independent methods in quantum
communication
but the very last i see
sarah you're nodding your head you

probably read this paper
one of the very last sentences in the
paper is
he says this redefines what physics is
about physics is about
languages and i think that's
a phenomenally profound statement i
think he's right
nice let's continue in order so we got
blue and then stephen and then sarah
so shanna's not here i think she said in
the chat that she um had to run
but um i want to touch back on what she
said like when
a thought passes that there's still
something there and also chris to what
you were saying earlier about
memory and what uh sarah was saying
of keeping track of what gets lost and
so something that i gleaned from this
paper now i have like no background
in category theory at all um as you may
have
been able to tell from the introduction
video right so um
the so it seems like that there's this
hierarchical nature of
some mathematical summation that is like
able to occur like via these methods and
i just was wondering if you're familiar
with
eric cole's work um he put out a paper
last year on causal geometry
and um like he's done some coarse
graining stuff
and so so is there some relationship
between like that kind of coarse
graining or

like dimension reduction and this
category theory work or is the category
theory
somehow able to encompass um the
entirety of the mathematical problem
in a better way or a different way than
say core screening
um yeah good good question
um uh yes i i do know eric's work
and i talk to him almost every week
because he's also part of
mike levin's lab and
so we have discussed this question of
of
course graining and causal emergence
and effective information all these
concepts that
that eric has been working with
um i think are very closely related to
the concepts that jim and i have been
working with
and we're
we're coming at them from such different
starting points that it's not clear
at least to me exactly how they fit
together
but what a what a coconut diagram does
is coarse-grain
it
it represents a
an abstraction
of a set of logical relationships
or or it represents a an abstractions of
a set of criteria into
a coarse-grained
larger scale if you will uh
criterion that
covers all of the cases below it

but in which information
is both lost and gained
i i think the the
the best example at least for me for
thinking about it is again going back to
cognitive categories if you think about
you know your category for a table or
another person or whatever
you you lose information in that
abstraction process
but you also gain information
by tying all of those exemplars
together and
each exemplar is then
in some sense encoding through the rest
of the network
of its consistency
in some sense with other exemplars that
may have
different uh detailed characteristics
so at the token level they look
inconsistent but at the tight level
they're consistent
and the
um the coco diagram represents
that that very complex relationship
so when
one is doing a course grading and it
works
the the information in the course grain
state
is capturing just this kind of abstract
information
it's the it's the abstract information
that's actually useful
and that captures what is consistent
between the
the fine-grained exemplars

and by organizing them that way it
allows each of them to represent
the fact of its consistency with this
larger set
thanks i also like i'm curious to see
because i do think that it's gonna
like you guys are we'll lock together
eventually at some point
um you know maybe in 10 years but but
i'm curious to see how that
um you know kind of dynamic interplay
unfolds
will also lie also um mike living will
be
on active inference live stream on july
13th
so we can talk before then or after then
all right awesome so we have um a little
bit less than 20 minutes
left so we'll just do you know questions
and try to get a few more questions in
from the chat as well
stephen and then sarah
thank you yeah deep breath there's a lot
a lot to
go through here um i was gonna sort of
build on that coarse grain
question and maybe just stepping back at
some of the bigger implications of
contextuality
um and how that is thought about so
um you know the principles that you
talked about or the paradigms
um so i'm curious about how you see this
maybe impacting contextuality
um and the way people think about that
um
let's address that and then continue

stephen let's just get one question and
one answer
okay
okay i guess i didn't i didn't actually
hear that as a question
sorry guys okay thank you sorry
yeah i just i just finished that thought
but basically
just in terms of contextuality and
coarse graining
and you know how much granularity is
thought about
um and how that has a bigger
that that general idea of how to think
about contextuality
um i'd be interested in your thought
about that as how that
sort of has a bigger set of legs and one
of the reasons i say that is because
when people often talk about quantum
theory
in a sort of broader world out there
they often think about this
transpersonal
it's all the same and it's unifying and
it's actually decontextualizing in some
ways you know it's almost
some ways this is quite revolutionary to
me to really see quantum
effects as being contextual you know
being in that way so you know that maybe
ties in with david bowman's work
on implicit order and the awareness
of things so rather than always being
about the whatness and going up
indigenous approaches are very much
about where you are and how you are
what's your embeddedness in your niche

so
in some ways it's like quantitative now
when i first heard about your paper i
was thinking oh this is
you hear quantum you think abstract but
actually
it's doing something very different i
just wondered how that might have impact
in other practices okay
so so i i heard a number of different
ideas here
that that are are parts of questions
one is to do with with course training
and contextuality
and you put your finger very precisely
on the danger of course grading
uh
and this this you can also state this in
terms of the frame problem
it's it's often the little details that
come back to bite you
and when you coarse grain you're
specifically getting rid of those little
details
so this is uh
goodness it it it's like the the
metaphor of the hammer and the nail
right if
any choice of reference frame is
effectively a choice of course grading
and choosing a reference frame choosing
a course greeting determines what you
can see
but that also determines what you can't
see
and it's what you can't see
that is responsible for the contextual
effects

that you can't avoid by
by by you know um
learning more about the context if
you've if you've arranged your
measurement procedures in a way that
prevents you from seeing them
then you can never take them into
account when you're building your
probability distribution
so so coarse graining
is something that we're stuck with
but it's it's very much a double-edged
sword
right it allows us to get around in the
world it allows us to see these
similarities
it allows us to learn things but
it forces us to ignore things that
may turn out to be important or may turn
out to have been
important
thanks for the really interesting answer
there
um so so with respect to you
to your other question about if you will
embeddedness
uh yes i think that's crucially
important
um let's go to sarah
i'm just going to say yeah that that
that whole embeddedness
is huge for the inactive ecological
cognition work and sort of the
the pragmatic turn in psychology so
there's a lot of implications of that so
i just
think that's very exciting so thanks
yeah

yeah i i'm very attracted to this sort
of
notion that ecosystems are
are in a sense extended organisms etc
that whole way of thinking in biology
is i think very important
and also to connect that it's like
there's this embedded practice and then
we're talking about the physical
electronic circuits
so it's almost like across domains we're
taking an
action-oriented and a realist manifested
focus but then seeing the what is
actually manifest as the tip of the
iceberg
in another sense but it's totally real
for what it really is
which is our measurement and then also
it's the tip of an iceberg because we're
also in the system and
those kinds of systems don't just arise
out of nowhere so
sarah and then scott in our last couple
of minutes
but thanks everyone for this really
awesome discussion
maybe sarah yeah good god
yeah um so am i to understand that we
have another one of these because i i
would love to ask chris
you know these other questions but
obviously not now there's just not
enough time
we will be discussing the paper next
week but um we did not plan without
totally
no pressure but we're doing the same

time next week but
no expectation okay no prior okay um
yeah in some ways this is like a media
um just rephrasing what's already been
said but it's actually right up right up
your alley steven
um i was actually just in writing in my
paper you know about well
what methods you choose depend on what
met what questions you ask
um related to reductionism related to
this whole theme
um but i i wonder this is like super
abstract
um but like i wonder about
the the role of play because i was
thinking about you know like
i was thinking about um there's so many
situations in science where you're not
you know like
when you look at the paper you're like
oh yeah they're doing science but in
reality they're just
poking at right like so it's just
this way of being with the world where
you're like oh i wonder what happens if
i you know but they're not really
reducing
they're just it's like so it's like play
and so i've really wondered about
about how to bring that to the
foreground about what that what that is
in science
you know and so i again i apologize but
you're
you always have a way of coming up with
a comment so hopefully you'll have a
thought on this

[Music]

oh yeah

if it's not fun you're probably doing it

wrong

i was just thinking of like an antidote

to reductionism like play seems like a

way that you're not

taking things apart you're just kind of

you're with it

in some way

well i to take this back to active

inference and to

to think of it uh

almost in in robotics terms right

how what the people in robotic and

developmental robotics are

are trying to do is understand

what motivates a system

to

engage with the world with this kind of

innocence that i think you're referring

to

to to just explore and

and see what's learnable

right the uh one of the one of the key

questions is

is um

and pierre etier is is the one who i

think in the literature has

emphasized this more than anyone else is

how do how do you learn what

parts of the world are learnable

with the resources that you have

so that you can avoid spending all of

your resources

trying to learn things that you're not

capable of learning

so how do you how do you how do you

focus your exploratory behavior
in the parts of the world where it can
actually be productive
this is sort of a meta question on top
of active inference
and it has to do with bayesian precision
and things like that
but it's your expectations about
learnability
and then there's a flip side of this in
evolutionary theory right the whole
idea that that what organisms are doing
in evolution
is trying to improve their evolvability
uh so they're they're they're trying to
increase if you will the flexibility of
their lineage
to cope with with whatever the
environment can throw at them
and this is very much i think what
what's happening in play as you put it
you're
you're trying to to engage with the
world in a way that
is wanting to know
what the possibilities for further
engagement are
thanks for the answer and speaking to
play it's almost like you start with
what you have the rubber band
the tin foil so you start with the
actual but then you you think about what
if
and you think about adjacencies even
categorical adjacencies
um so that was a pretty interesting
so and then also a comment on what you
said about the expectations of

learnability
in active inference lab we would hope to
learn by doing and applying
and update people's expectations to
increase their um
understanding of they can learn active
inference basically
we'll figure out other ways to say it
more directly but
will manifest through observations on
people's screens
that they can learn active inference and
then it's almost like in the cognitive
dimension
our collective and our individual
cognitive evolution will
undertake another inflection point when
we believe that we can learn
so that will be really powerful so scott
yeah along those lines just in terms of
further leverage i want to talk a little
bit about the buy
or build decision that every organism
makes and the idea is do you do you
develop it internally or do you get it
outside right
and so it seems like one of the things
that struck me is we've talked about the
brain
being tied to language one of the things
i've been
fussing with lately and i've this is the
question i've been
fussing with the idea that the mind
resides in language and the brain is
just an
antenna tuned to the mind
now the antenna would be the markov

blanket right or the
the action of the markov blanket is a
continuous dynamic tuning
and the so that you're if you're born
into a
cultural context in my place very
quickly my sister adopted a
five-month-old chinese baby she was
raised in central pennsylvania she looks
chinese she
doesn't speak chinese doesn't like
chinese food she's not chinese right
so her mind is a creature of her
environment
in body so one of the things in terms of
affordances you know we talk about
resources internally to develop the
markov like it
you know you can really shortcut that if
you say hey
i know that my resource internally is i
should time myself to
to daniel because he's awesome and he
knows all the stuff i don't have to know
it i just have to know if i have a
question i'll go ask him
right so you start to have something
like it's like eukaryotic revolution and
multicellularity
you start to have this de-risking and
leverage at
larger scales made possible by the fact
of synthesis of multiple markov blankets
now
they can be synthesized around common
ideas or common problems
so the the question is in terms of that
again

separateness we see ourselves and our
markov blankets and our minds as
separate
but it seems like maybe they're not
maybe maybe the mind is
all one thing and we have these
instantiations of it that we call
different
there are different iterations of that
markov language can you comment on that
kind of macro and micro level uh notion
briefly um
i'll try i think this is all good
thinking
i i view this in the language of
in a sense extended organism thinking
and evolutionary theory
which is very closely connected to
to multicellularity and evolutionary
theory
uh which is closely connected to
the sorts of of
faculty multicellularity you see in
microbial communities
um i think from the
from i think we see examples of this
throughout biology
and um so the question becomes how do
how do we take this into the human
sphere or
as i think you posed it very clearly uh
how do we realize that it's already been
taken into the human sphere
we just don't quite get it yet
and and i suspect that you're right
uh that we just don't quite get it yet
and you know i i remember in graduate
school someone said

you're what you're learning is how to
read the literature yes
you don't need to learn any facts
and that has to do with security right
and in new forms of
threat and integrity and privacy issues
all these new allergic
problems are not necessarily
new problems okay and that's why i
wondered about the
biology and the biomimicry you see that
we can
how can we actively pursue biomimicry
thank you cool sorry you're a little
quiet at the end there
scott but um with this um close the hour
um chris really thanks so much and also
to
um shauna for joining with your
expertise but really this was a special
event for us we would always welcome the
opportunity to
speak with you or colleagues to learn
more
and any other final thoughts
um that you'd like to give in closing
chris
uh yeah feel free to send me an email
my my address is on my website
chrisfieldresearch.com
all the papers i published recently are
there um
but yeah i'd love to continue discussing
awesome well thanks a lot and um next
week we're going to be having a
follow-up discussion to really
digest this definitely re-listen to it
and go from there so thanks everyone for

joining and we'll see you

another time